CURRICULUM VITAE OF FRANCESCO PIO MARINO
November 2026

Personal Data

CONTACTS

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Education

Ph.D. in Computer Science 08 2023 | 09 2026
University of Rouen Normandy, France
Supervisors: Prof. Simone Faro and Prof. Thierry Lecroq.
Thesis: Text Sampling for Efficient Information Processing.

Ph.D. in Computer Science 08 2023 | Present
University of Catania, Italy
Supervisors: Prof. Simone Faro and Prof. Thierry Lecroq.
Thesis: Text Sampling for Efficient Information Processing.

Erasmus Exchange 09 2021 | 02 2022
Kaunas University of Technology, Lithuania

Master's Degree in Computer Science 09 2020 | 07 2022
University of Catania, Italy

Visiting Research Positions

Universidade Estadual de Campinas (UNICAMP), Brazil 02 2026 | 03 2026
Visiting researcher for scientific collaboration with **Prof. Guilherme P. Telles** and **Prof. Felipe A. Louza**; research on indeterminate strings and indexing.

University of Yamanashi, Japan 09 2025 | 12 2025
Visiting researcher for scientific collaboration with **Prof. Dominik Köppl**;
research on string algorithms, compressed data structures, and recompression.

Scientific and Research Activity

RESEARCH PROFILE

Computer Science researcher specializing in string algorithms and data compression, with research spanning exact and approximate pattern matching, text sampling, compressed indexing, indeterminate and elastic-degenerate strings, and quantum algorithms. International research experience in France, Japan, and Brazil.

RESEARCH INTERESTS

String algorithms, exact and approximate pattern matching, data compression, compressed text indexing, string sampling, algorithms on indeterminate and elastic-degenerate strings, combinatorial algorithms, quantum algorithms.

TEACHING & TUTORING

- **Junior Tutor, University of Catania** (2019–2025): preparation for the Italian Olympiad in Informatics (2019–2025).
- **Junior Tutor, University of Catania**: Pink Coding (2021); remedial tutoring in Computer Science (2024–2025).

SELECTED CONFERENCE PRESENTATIONS

- *Prague Stringology Conference 2020 (PSC20)*, 31 August–2 September 2020, Prague, Czech Republic.
- *Workshop on Compression, Text and Algorithms (WCTA 2020)*, 16 October 2020, Online.
- *Prague Stringology Conference 2021 (PSC21)*, 30–31 August 2021, Prague, Czech Republic.
- *Italian Conference on Theoretical Computer Science 2021 (ICTCS21)*, 13–15 September 2021, Bologna, Italy.
- *Quantum Search and Information Retrieval (QUASAR24)*, 3 June 2024, Pisa, Italy.
- *12th International Conference on Fun with Algorithms (FUN24)*, 4–8 June 2024, La Maddalena, Italy.
- *Future Technologies Conference (FTC24)*, 14–15 November 2024, London, United Kingdom.
- *29th London Stringology Days & London Algorithmic Workshop (LSD & LAW)*, 5 February 2025, London, United Kingdom.

- *Sequences in London*, 6–7 February 2025, London, United Kingdom.
- *2nd Workshop on Quantum Algorithms, Software and Applied Research (QUASAR25)*, 20 July 2025, South Bend, IN, USA.
- *Data Compression Conference 2026 (DCC26)*, 24–27 March 2026, Snowbird, UT, USA.
- *Combinatorial Pattern Matching 2026 (CPM26)*, 15–17 June 2026, Copenhagen, Denmark.
- *3rd Workshop on Quantum Algorithms, Software and Applied Research (QUASAR26)*, 13 July 2026, Cleveland, OH, USA.
- *27th Italian Conference on Theoretical Computer Science (ICTCS26)*, 7–9 September 2026, Udine, Italy.

INVITED SEMINARS

- *Enhancing Exact and Approximate String Matching by Character Distance Sampling*, University of Rouen Normandy, 28 March 2024.
- *From Elastic-Degenerate Strings to Full-Text Indexing: Suffix Sorting, BWT, FM-Index Construction and Search*, University of Caen Normandy, 6 May 2026.

ACADEMIC SERVICE

- **Program Committee Chair**, *QUASAR 2026*, 3rd Workshop on Quantum Algorithms, Software and Applied Research, 13 July 2026, Cleveland, OH, USA.
- **Organizer**, String-Masters Workshop, 4 December 2025, Kofu, Japan.
- **Organizer**, Catania String Days, 19–20 January 2026, Catania, Italy.

REFEREE SERVICES

Reviewer for the following international journals and conferences:

- ACM Transactions on Quantum Computing
- Frontiers in Quantum Science and Technology
- Discover Computing
- STACS 2027
- WALCOM 2027
- Prague Stringology Conference 2026 (PSC26)
- QUASAR 2026

AWARDS

- *Best Student Paper Award* – 3rd Workshop on Quantum Algorithms, Software and Applied Research (QUASAR 2026). *Paper: Delta-Based Rare Pattern Discovery with Quantum Speedup.*

GRANTS & FUNDING

- *INdAM–GNCS Project 2025: STaRQ – Soluzioni per l'Analisi e la Ricerca Quantistica di Testi. Role: Member* *EUR 1,500*
- *French Embassy in Italy – Cassini Junior: Catania String Days. Role: Principal Investigator* *EUR 500*
- *LITIS – Appel à projets scientifiques 2025: Japan PhD Mobility Grant. Role: Principal Investigator* *EUR 2,500*

List of Scientific Papers

INTERNATIONAL JOURNALS (7 PAPERS)

1. Simone Faro, Thierry Lecroq, Francesco Pio Marino, Arianna Pavone and Antonio Scardace. Parameterized String Matching by Text Sampling. **Mathematics**, 14(15):2743, 2026. doi:10.3390/math14152743.
2. Francesco Pio Marino and Jorma Tarhio. Improving String Matching through Enhanced Hashing. **Software: Practice and Experience**, 56(11):1323–1333, 2026. doi:10.1002/spe.70087.
3. Simone Faro, Thierry Lecroq, Francesco Pio Marino, Antonio Scardace and Arianna Pavone. Monotonic Run Length Text Sampling and Fake Sampled Positions. **The Computer Journal**, 2026. doi:10.1093/comjnl/bxag075.
4. Simone Faro, Thierry Lecroq and Francesco Pio Marino. Optimal Character Distance Sampling for Exact String Matching through Set Cover Reformulation. **IEEE Access**, 14, 2026. doi:10.1109/ACCESS.2026.3654667.
5. Simone Faro, Francesco Pio Marino and Gabriele Messina. Extending Qutes: A Practical High-Level Language for Quantum Computing. **The Computer Journal**, 69(3):597–612, 2026. doi:10.1093/comjnl/bxaf133.
6. Simone Faro, Francesco Pio Marino and Arianna Pavone. Improved Characters Distance Sampling for Online and Offline Text Searching. **Theoretical Computer Science**, 946:113684, 2023. doi:10.1016/j.tcs.2022.12.034.
7. Simone Faro, Francesco Pio Marino and Arianna Pavone. Efficient Online String Matching Based on Characters Distance Text Sampling. **Algorithmica**, 82(11):3390–3412, 2020. doi:10.1007/s00453-020-00732-4.

INTERNATIONAL CONFERENCES (22 PAPERS)

8. Daniele Lizzio Bosco, Jacopo Cossio, Simone Faro, Francesco Pio Marino, Gabriele Messina, Carla Piazza and Giuseppe Serra. AI-Assisted Discovery of CNOT Decompositions for Quantum Permutations on Constrained Topologies. **Proceedings of the 4th International Workshop on AI for Quantum and Quantum for AI (AIQxQIA 2026)**.

9. Simone Faro, Francesco Pio Marino and Arianna Pavone. Shallow Quantum Shuffling via Measurement-Driven Multi-Scale Rotations. **Proceedings of the 27th Italian Conference on Theoretical Computer Science (ICTCS26)**.
10. Dominik Köppl and Francesco Pio Marino. Recompressing Compressed Data. **Proceedings of the 27th Italian Conference on Theoretical Computer Science (ICTCS26)**.
11. Thierry Lecroq, Arnaud Lefebvre and Francesco Pio Marino. Computation of Periods and Covers Using Sampling. **Proceedings of the Prague Stringology Conference 2026 (PSC26)**.
12. Simone Faro, Farida Farsian, Francesco Pio Marino, Gabriele Messina, Francesco Schillirò, Eva Sciacca and Fabio Vitello. Delta-Based Rare Pattern Discovery with Quantum Speedup. **ACM Symposium on High-Performance Parallel and Distributed Computing 2026 (HPDC26)**.
13. Simone Faro, Francesco Pio Marino and Gabriele Messina. Structural Parallelism in Quantum Programs. **ACM Symposium on High-Performance Parallel and Distributed Computing 2026 (HPDC26)**.
14. Simone Faro, Francesco Pio Marino, Arianna Pavone, Simone Spina and Caterina Viola. Quantum Bit-Parallel Swap Matching. **ACM Symposium on High-Performance Parallel and Distributed Computing 2026 (HPDC26)**.
15. Simone Faro, Francesco Pio Marino and Gabriele Messina. Spells for Quantum Programmers: Expressive High-Level Commands in Qutes. **Fun with Algorithms 2026 (FUN26)**.
16. Simone Faro, Dominik Köppl, Thierry Lecroq and Francesco Pio Marino. A Bitwise Approach to SCER Matching in Indeterminate Strings. **Combinatorial Pattern Matching 2026 (CPM26)**.
17. Simone Faro, Dominik Köppl and Francesco Pio Marino. Attractor Matching: A New Paradigm for Structural String Comparison. **Data Compression Conference 2026 (DCC26)**.
18. Simone Faro, Dominik Köppl, Thierry Lecroq and Francesco Pio Marino. Enabling FM-Index for Elastic-Degenerate Strings via a New Min/Max Wavelet Tree. **Data Compression Conference 2026 (DCC26)**.
19. Simone Faro and Francesco Pio Marino. Scaling Grover's Search for Large Solution Spaces. **ACM Symposium on High-Performance Parallel and Distributed Computing 2025 (HPDC25)**.
20. Simone Faro, Francesco Pio Marino and Gabriele Messina. Qutes: A High-Level Quantum Programming Language for Simplified Quantum Computing. **ACM Symposium on High-Performance Parallel and Distributed Computing 2025 (HPDC25)**.
21. Simone Faro, Francesco Pio Marino, Andrea Moschetto, Arianna Pavone and Antonio Scardace. The Great Textual Hoax: Boosting Sampled String Matching with Fake Samples. **Proceedings of the 12th International Conference on Fun with Algorithms (FUN24)**.
22. Simone Faro, Francesco Pio Marino and Antonio Scardace. Practical Implementation of a Quantum String Matching Algorithm. **Quantum Search and Information Retrieval (QUASAR24)**.
23. Simone Faro, Francesco Pio Marino and Andrea Moschetto. Beyond Horspool: A Comparative Analysis in Sampled Matching. **Proceedings of the Prague Stringology Conference 2024 (PSC24)**.
24. Domenico Cantone, Claudio Caudullo, Simone Faro, Francesco Pio Marino, Arianna Pavone and Caterina Viola. Practical Quantum Combinatorial String Matching. **Future Technologies Conference (FTC24)**.
25. Simone Faro, Thierry Lecroq, Francesco Pio Marino, Arianna Pavone and Stefano Scafiti. Improving Sampled Matching through Character Context Sampling. **Proceedings of the 25th Italian Conference on Theoretical Computer Science (ICTCS24)**.
26. Simone Faro, Francesco Pio Marino, Arianna Pavone and Antonio Scardace. Towards an Efficient Text Sampling Approach for Exact and Approximate Matching. **Proceedings of the Prague Stringology Conference 2021 (PSC21)**.
27. Simone Faro, Francesco Pio Marino and Arianna Pavone. Enhancing Characters Distance Text Sampling by Condensed Alphabets. **Proceedings of the 22nd Italian Conference on Theoretical Computer Science (ICTCS21)**.
28. Simone Faro and Francesco Pio Marino. Reducing Time and Space in Indexed String Matching by Characters Distance Text Sampling. **Proceedings of the Prague Stringology Conference 2020 (PSC20)**.
29. Simone Faro, Francesco Pio Marino and Stefano Scafiti. Fast-Insertion-Sort: A New Family of Efficient Variants of the Insertion-Sort Algorithm. **Proceedings of the SOFSEM 2020 Doctoral Student Research Forum, co-located with the 46th International Conference on Current Trends in Theory and Practice of Computer Science (SOFSEM20)**.